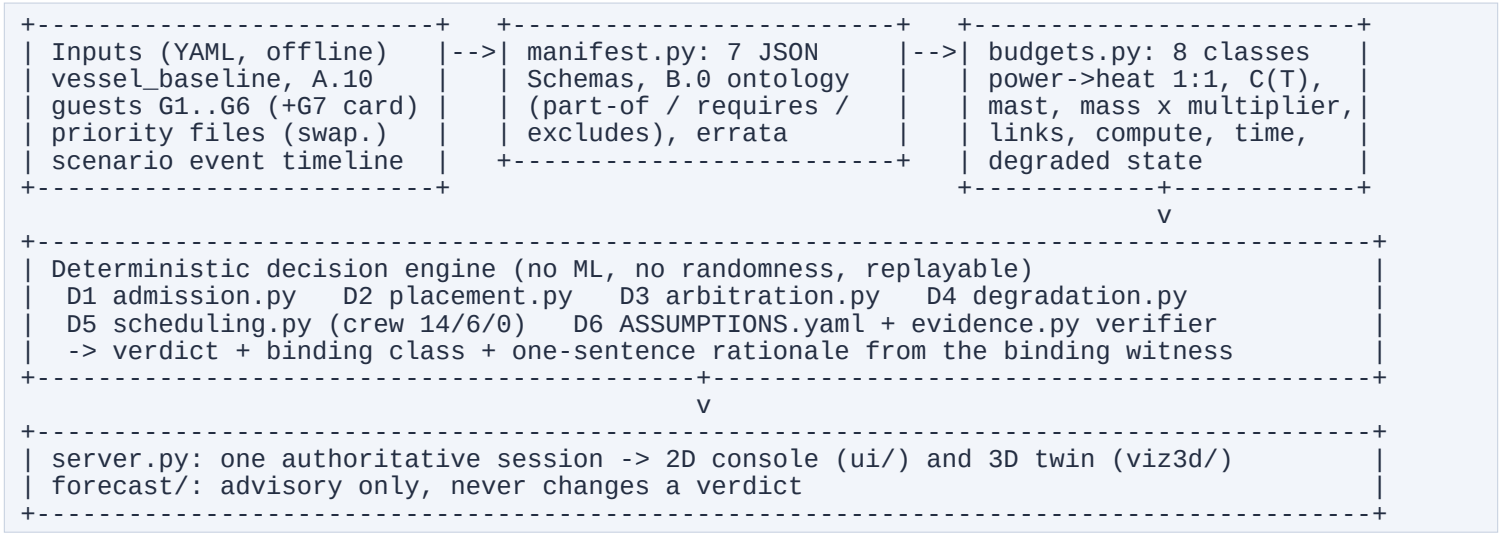
SDV Resource Governor (sdv-resource-governor)

Run the demo (one command): ./run_demo.sh opens the operator console at http://127.0.0.1:8765 (creates .venv and installs on first run). The same governor runs on the Arduino Ventuno Q (http://10.0.0.130:8765, trusted LAN only: the console is unauthenticated).

A deterministic resource governor for the 60-70 m reference vessel: it owns the eight Annex A budgets and decides admission (D1), placement (D2), right of way (D3) and degradation (D4), treats crew (D5: 14 / 6 / 0) as a setting, and keeps every assumed number in the Physical Account (D6). Every verdict names its binding constraint in one sentence built from the witness that bound it.

1. Architecture Diagram & Data Flow



2. Deliverables, Objectives & Scope (D1-D6)

ID	Decision	What it optimises (declared objective)	Component and result
D1	Admission in 30 s (mandatory)	Feasibility over the whole 14-day A.10 envelope, degraded states included, never the pier; YES_IF displacements ranked by least capability lost, then fewest guests; states whether the verdict holds with one genset down (YES / YES_WITH_STEP_DOWN / NO).	admission.py: YES / YES_IF / NOT_NOW / NO with binding class. At sea with 0 crew it adds up to 3 logical reconfigurations (reconfiguration.py).
D2	Placement (addressed)	Objective D2_LEXICOGRAPHIC_V2, key (roof_cond_rank, pref_penalty, -retention_milli, scarcity_cost, -min_margin_bucket, stab_eq, node_used_cores, nid): avoid conditional roof positions, then preferred positions, capability retained, least scarce slot (free mast = 3), largest minimum margin, least stability-equivalent mass, least-loaded node.	placement.py: bay, mast, deck, power line, cooling manifold; up to 5 ranked placement_alternatives with why_not_chosen and 5 rejected_placements with their binding witness.
D3	Right of way in 2 s (mandatory)	Max utility_score = priority-weighted survey continuity + priority-weighted landing margin, subject to the hard G4 < 20 % battery deadline; tie-break declared in the priority file.	arbitration.py: survey_primary lands G4 at T+1095 s (105 s margin, G3 loses 14 min); rescue_primary at T+855 s (345 s margin, G3 loses 26 min).
D4	Degradation (addressed)	Restore power/cooling feasibility with least capability lost: step down before shedding, in the priority file's order; never shed safety-critical, uninterruptible, airborne or submerged guests.	degradation.py: genset trip at T=720 s, committed 106.5 > 100 kW, G5 45 -> 8 kW; rationale states what every guest keeps or loses.
D5	Crew as a setting (memo)	One rule, "nobody on the flight deck during recovery", with crew-dependent confirmation (30 / 120 s / none) and deck clearing (240 / 600 s / impossible).	scheduling.py, arbitration.py; docs/MEMO_SCENARIO_C.pdf.
D6	Physical Account	Every assumed number traceable to the code that uses it.	docs/PHYSICAL_ACCOUNT.pdf, ASSUMPTIONS.yaml, evidence verifier (28 mutations, 0 false accepts).

3. Quickstart (offline, < 5 minutes)

```
python3 -m venv .venv --system-site-packages
.venv/bin/pip install -r requirements-macos-arm64.lock
.venv/bin/pip install --no-build-isolation --no-deps -e . && .venv/bin/pytest -q
.venv/bin/sdv-governor campaign --run-id demo_run
.venv/bin/sdv-governor verify-evidence artifacts/demo_run --run-mutations
.venv/bin/sdv-governor serve --host 127.0.0.1 --port 8765
```

4. Eight-Class Resource Model

Class	Model (Scenario B values)	Module

Power	200 kW reserve (100 kW after a trip); two 150 kW lines; committed vs actual (106.5 -> 69.5 kW)	budgets.py
Cooling	C(T) 160 -> 120 kW from 12 to 30 C (133.3 kW at 24 C); -40 % loop fault; inertia 30 kW x 60 s	budgets.py
Positions	bays, flight deck / ramp exclusion, mast_3 (60 kg, 2,000 W air), hangar roof under a recovery condition	budgets.py, placement.py
Mass	10,000 kg and 12,000 kg-eq; G1+G3+G4+G5 = 9,560 kg / 9,853 kg-eq	budgets.py
Data links	SATCOM 2-8 Mbit/s (A.5; 6 in A.10 cruise, 0 in silence); acoustic link down with layered water, manoeuvre or deck machinery	budgets.py, acoustic.py
Compute	4 nodes x 32 cores, 80 % allocatable	placement.py
Crew and deck	14 / 6 / 0: confirm 30 / 120 s / none; deck clear 240 / 600 s / impossible	scheduling.py
Time	30 s / 2 s / 5 s / 60 s checked every run; A.6 transition times; safety command < 1 ms measured	replay.py

5. Adding a 7th Guest Card Without Touching Code

Write a YAML card with the B.0 fields (consumption per mode, positions, part-of / requires / excludes, conditions, installation, capability) as in docs/examples/G7_xbt_launcher.yaml, then parse and admit it:

```
import yaml
from pathlib import Path
import sdv_governor
from sdv_governor.manifest import parse_guest_manifest

repo = Path(sdv_governor.__file__).resolve().parents[2]
card = repo / "docs/examples/G7_xbt_launcher.yaml"
raw = yaml.safe_load(card.read_text(encoding="utf-8"))
manifest = parse_guest_manifest(raw)  # fail-closed: unknown field, bool as number
```

Pass manifest to admission.evaluate_admission (or drop the file in data/guests/ to make it part of the catalog). A craft that launches or recovers sets launch_recovery_area (flight_deck / stern_ramp) and inherits the A.3 deck/ramp exclusion; a mode that needs the acoustic link sets requires_acoustic_command_link: true.

6. Explicit Limitations & Honest Operational Boundaries

* 1. The < 1 ms safety-command latency is the governor’s own state change and command emission (budget 200 ms); breakers, winch brakes, the acoustic modem (15 s modelled) and the turn into wind (90 s modelled) are physical plant outside our control.
* 2. Scenario B runs in 20 s ticks with piecewise-linear cooling and one lumped thermal reservoir, not a pipe-network model.
* 3. Telemetry is synthetic; public metocean data is a recorded offline snapshot. The live 90 s human rehearsal is still to be done by the owner.
* 4. Crew timings are single mid-range values from A.6, not distributions.

7. What We Would Do With One More Week

* 1. Hardware-in-the-loop: feed telemetry and commands over Modbus TCP / NMEA 2000 with watchdog heartbeats.
* 2. Per-bay chilled-water model and generator droop during the 35 kW drone charge.
* 3. Crew timings as distributions, and the Scenario C rule checked against a rolling 72 h weather forecast so recoveries are planned before the deck closes.
* 4. Signed audit log of every decision and operator override.

8. Public Data & Forecasting (advisory)

Recorded Golden Gate metocean data (Open-Meteo, NOAA NDBC 46237) drive SST, sea state and wind in replays and guarded 48 h window outlooks. A damped Holt forecaster with conformal bands warns before a guest overruns (median lead 60.5 s; p99 tick 2.68 ms on Ventuno Q); it never changes a verdict. Details and gates: docs/FORECAST_AND_PUBLIC_DATA.pdf.

9. Repository Structure

src/sdv_governor/ engine; data/ and schemas/ ship, guests, priorities, scenarios; ui/, viz3d/ operator screens; docs/ deliverables (Markdown + PDF); tests/, artifacts/ test suites and evidence bundles.